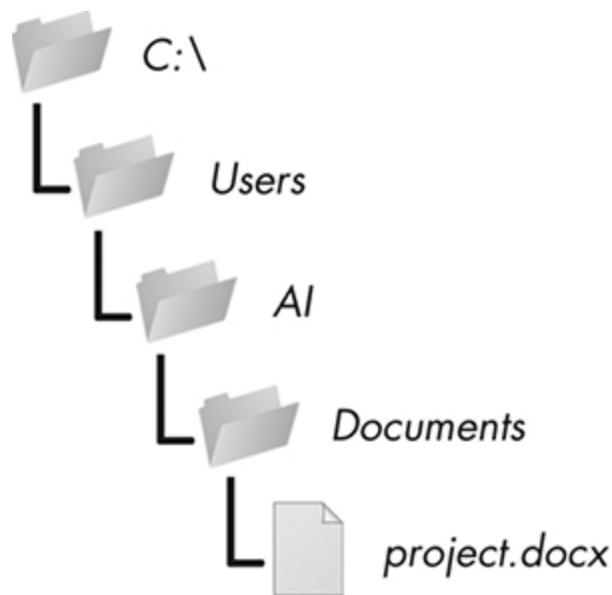


settings to get your system working well enough might hide problems, but it won't fix them. By taking the time to understand these issues now, you can prevent them from reoccurring.

The Filesystem

The *filesystem* is how your operating system organizes data to be stored and retrieved. A file has two key properties: a *filename* (usually written as one word) and a *path*. The path specifies the location of a file on the computer. For example, a file on my Windows 10 laptop has the filename *project.docx* in the path *C:\Users\AI\Documents*. The part of the filename after the last period is the file's *extension* and tells you a file's type. The filename *project.docx* is a Word document, and *Users*, *AI*, and *Documents* all refer to *folders* (also called *directories*). Folders can contain files and other folders. For example, *project.docx* is in the *Documents* folder, which is in the *AI* folder, which is in the *Users* folder. [Figure 2-1](#) shows this folder organization.



[Figure 2-1](#): A file in a hierarchy of folders

The *C:* part of the path is the *root folder*, which contains all other folders. On Windows, the root folder is named *C:* and is also called the *C:* drive. On macOS and Linux, the root folder is */*. In this book, I'll use the